

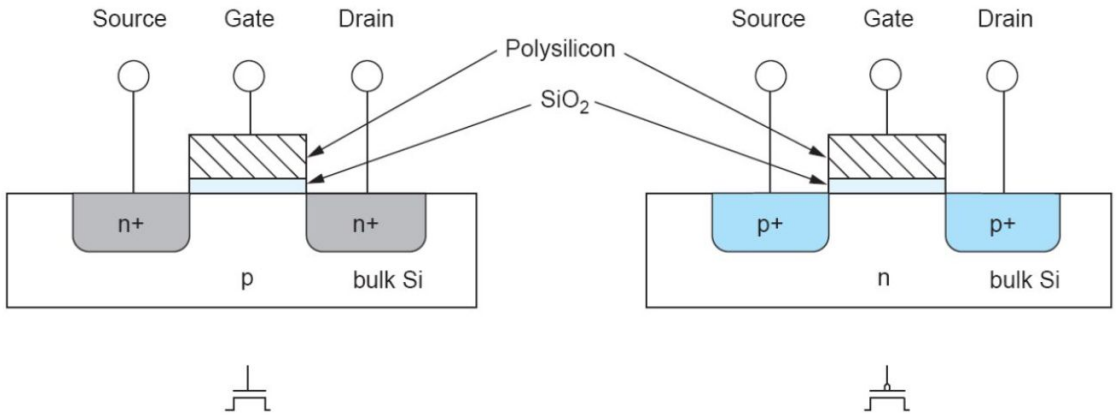
Universidade de Brasília – UnB
Faculdade de Ciências e Tecnologias em Engenharia

Digital Integrated Circuit Design

CMOS Inverter

MOS Transistor

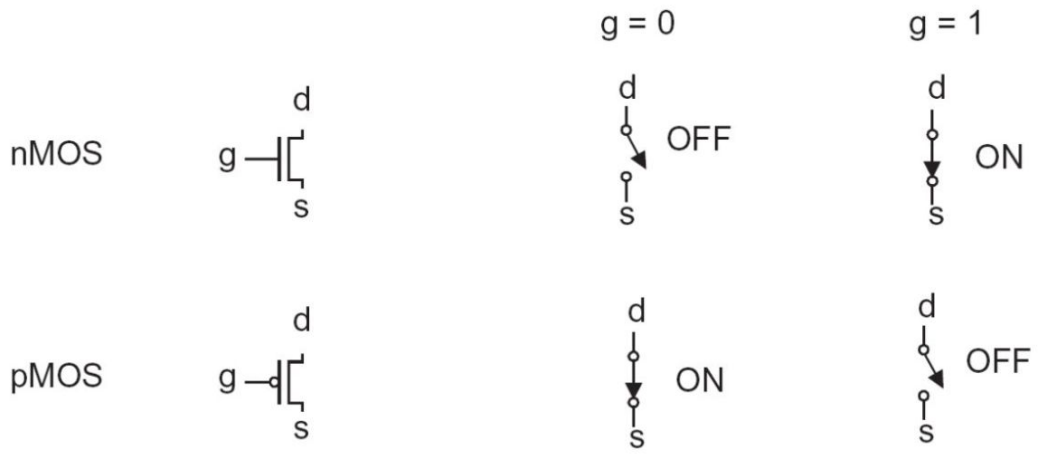
- The gate is a control input: gate affects the flow of electrical current between the source and drain
- NMOS: Vgs > 0 creates an electric field that starts to attract free electrons under the gate and creates the channel - region acts as an n-type semiconductor
- PMOS is reversed



Weste, N. H. E., Harris, D. M., ***CMOS VLSI Design: A Circuits and Systems Perspective***, 4th ed., Addison-Wesley, 2011.

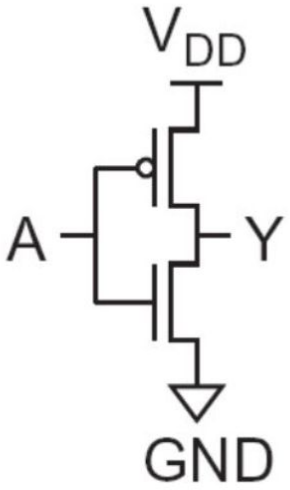
Switch-Level Model

- Vdd = logic 1; Gnd (Vss) = logic 0
- NMOS/PMOS
  - Vgate = 1: ON/OFF
  - Vgate = 0: OFF/ON

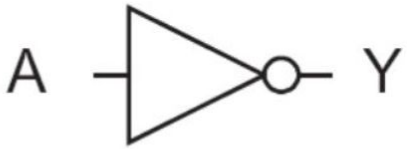


Weste, N. H. E., Harris, D. M., ***CMOS VLSI Design: A Circuits and Systems Perspective***, 4th ed., Addison-Wesley, 2011.

CMOS Logic Inverter



(a)



(b)

A= 0 -> NMOS is OFF; PMOS is ON

Y=1 (connected to Vdd)

A= 1 -> NMOS is ON; PMOS is OFF

Y=0 (connected to Gnd)

Table 1.1 Inverter truth table

A	Y
0	1
1	0

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DC Transfer Characteristics

$$V_{gsn} = V_{in}$$
$$V_{dsn} = V_{out}$$
$$V_{gsp} = V_{in} - V_{DD}$$
$$V_{dsp} = V_{out} - V_{DD}$$

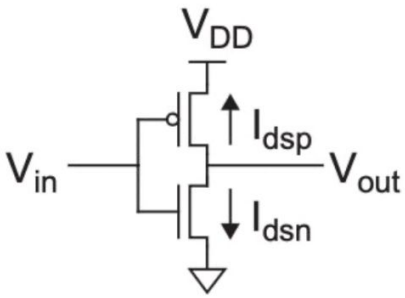


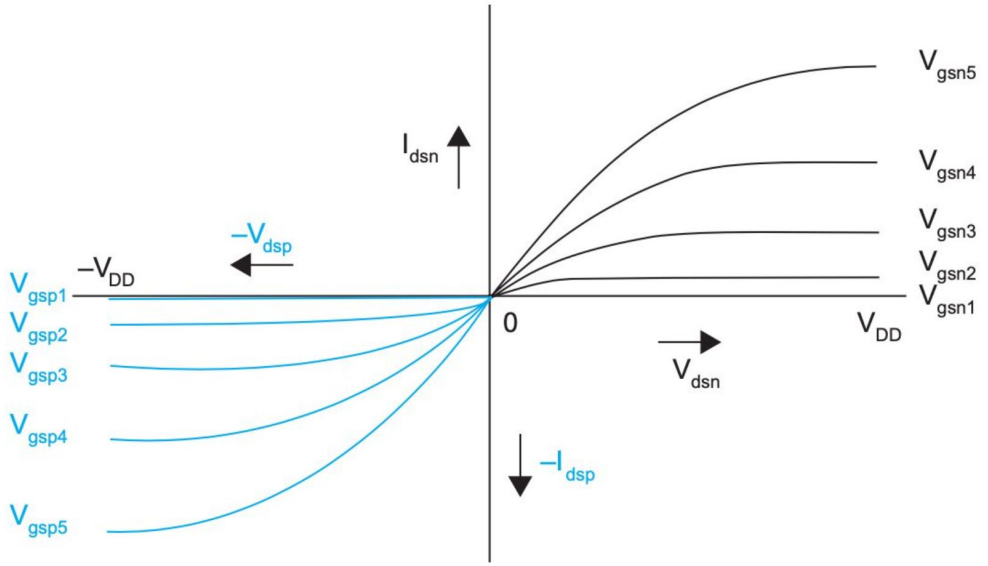
TABLE 2.2 Relationships between voltages for the three regions of operation of a CMOS inverter

	Cutoff	Linear	Saturated
nMOS	$V_{gsn} < V_{tn}$	$V_{gsn} > V_{tn}$	$V_{gsn} > V_{tn}$
	$V_{in} < V_{tn}$	$V_{in} > V_{tn}$	$V_{in} > V_{tn}$
		$V_{dsn} < V_{gsn} - V_{tn}$	$V_{dsn} > V_{gsn} - V_{tn}$
		$V_{out} < V_{in} - V_{tn}$	$V_{out} > V_{in} - V_{tn}$
pMOS	$V_{gsp} > V_{tp}$	$V_{gsp} < V_{tp}$	$V_{gsp} < V_{tp}$
	$V_{in} > V_{tp} + V_{DD}$	$V_{in} < V_{tp} + V_{DD}$	$V_{in} < V_{tp} + V_{DD}$
		$V_{dsp} > V_{gsp} - V_{tp}$	$V_{dsp} < V_{gsp} - V_{tp}$
		$V_{out} > V_{in} - V_{tp}$	$V_{out} < V_{in} - V_{tp}$

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DC Transfer Characteristics

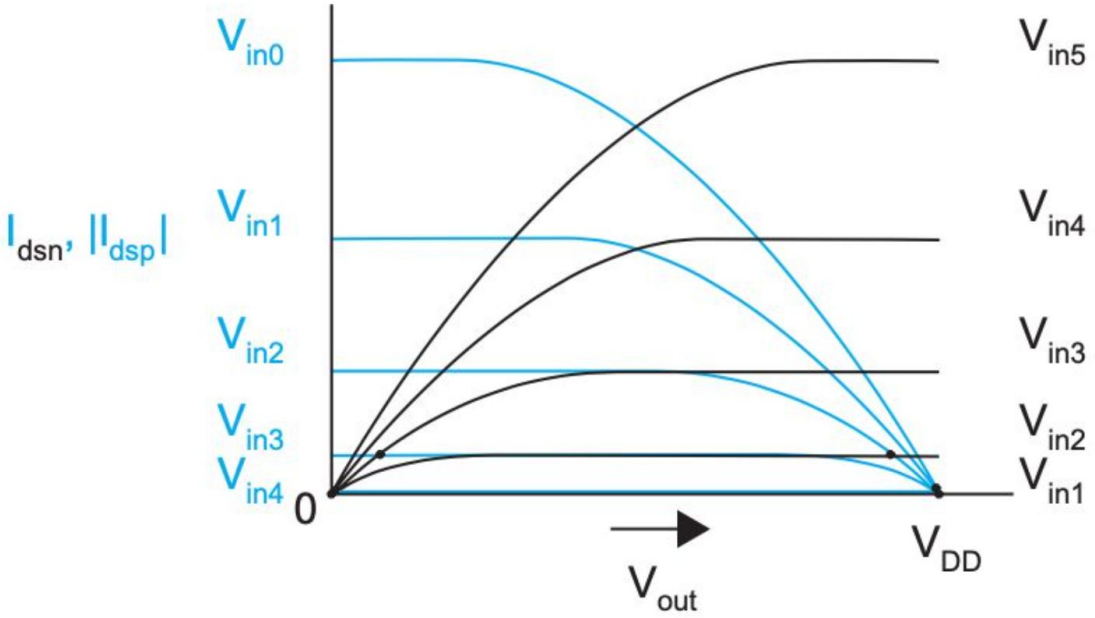
- Objective: find Vout x Vin
- $I_{dsn} = |I_{dsp}|$
- $V_{tp} = - V_{tn}$
- PMOS 2 - 3 times as wide as the NMOS ($\beta_n = \beta_p$)



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DC Transfer Characteristics

- I_{dsn} and $|I_{dsp}|$ in terms of Vout for various values of Vin
- Operating points: values of Vout where $I_{dsn} = |I_{dsp}|$ for a given Vin

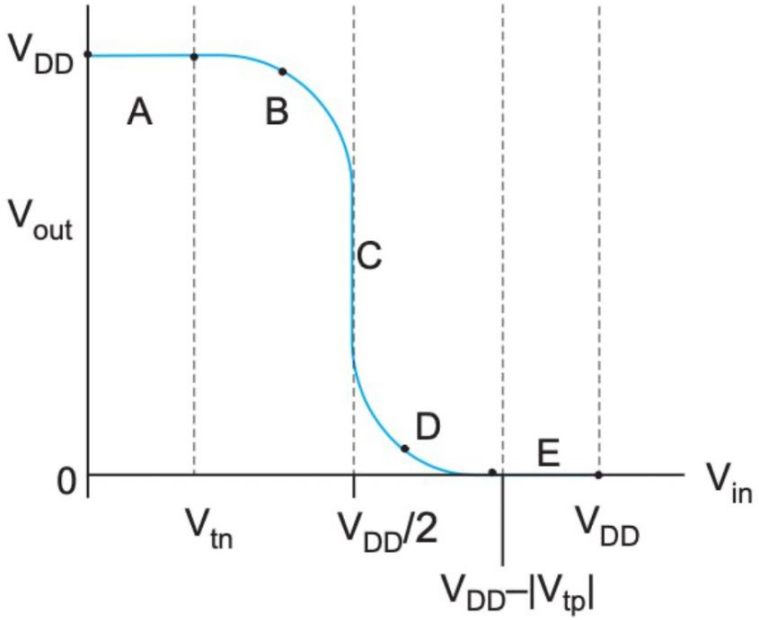


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DC Transfer Characteristics

TABLE 2.3 Summary of CMOS inverter operation

Region	Condition	p-device	n-device	Output
A	$0 \le V_{in} < V_{tn}$	linear	cutoff	$V_{out} = V_{DD}$
B	$V_{tn} \le V_{in} < V_{DD}/2$	linear	saturated	$V_{out} > V_{DD}/2$
C	$V_{in} = V_{DD}/2$	saturated	saturated	V_{out} drops sharply
D	$V_{DD}/2 < V_{in} \le V_{DD} - \lvert V_{tp}\rvert$	saturated	linear	$V_{out} < V_{DD}/2$
E	$V_{in} > V_{DD} - \lvert V_{tp}\rvert$	cutoff	linear	$V_{out} = 0$



Input threshold:

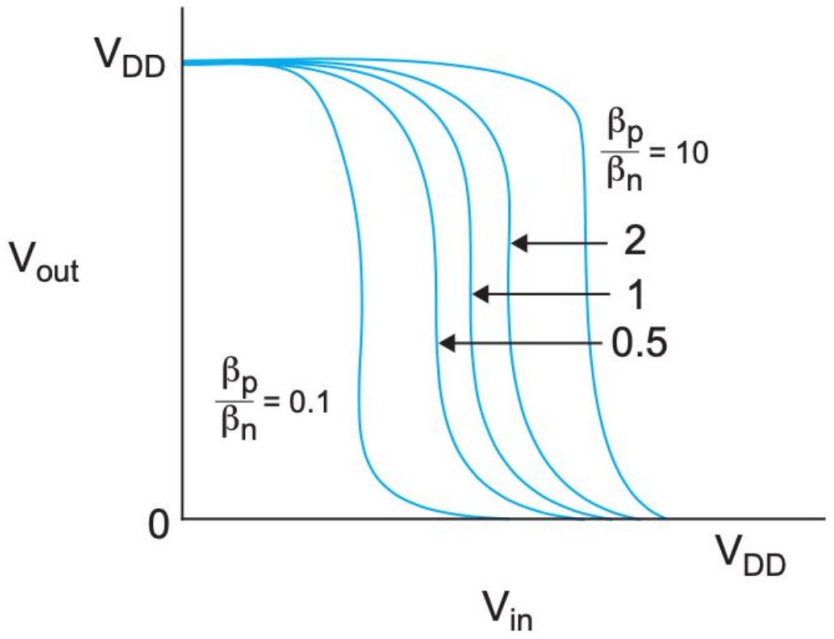
$$V_{inv} = V_{in} = V_{out}$$

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Beta Ratio Effects

- For $\beta_n = \beta_p$ the inverter threshold is Vdd/2: maximize noise margins
- $r = \beta_n / \beta_p$
- $r > 1$: HI-skewed
- $r < 1$: LO-skewed
- $r = 1$: unskewed

$$\beta = \mu C_{ox} \frac{W}{L}$$

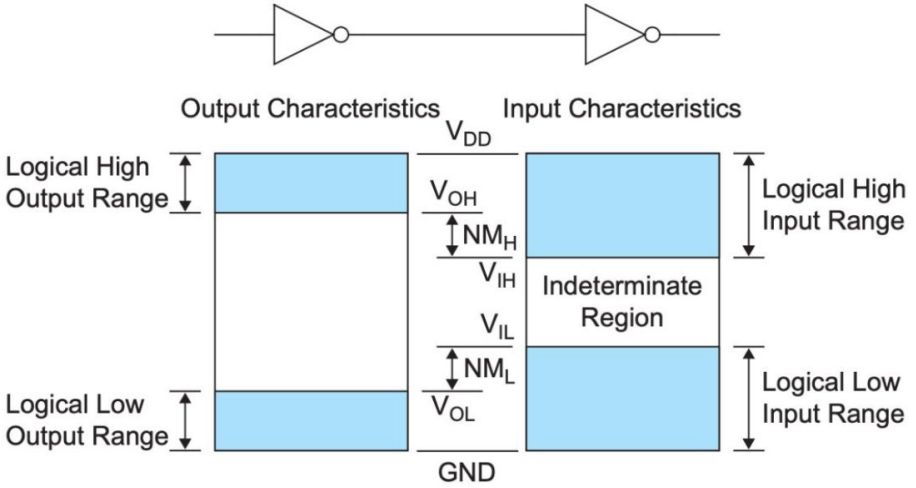


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Noise Margins

- Allowable noise voltage on the input of a gate so that the output will not be corrupted
- NM_L (low noise margin) = V_{IL} - V_{OL}
- NM_H (high noise margin) = V_{OH} - V_{IH}

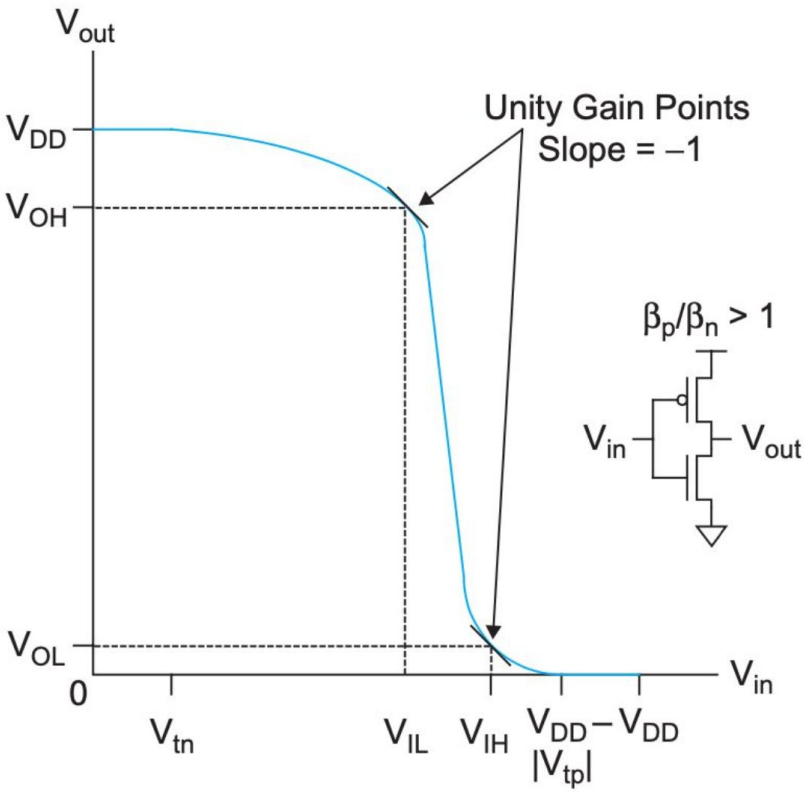
V_{IH} = minimum HIGH input voltage
V_{IL} = maximum LOW input voltage
V_{OH} = minimum HIGH output voltage
V_{OL} = maximum LOW output voltage



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Noise Margins

- Logic levels are defined at the unit gain point where the slope = -1
- If NM_L or NM_H are too small, the gate may be disturbed by noise in the inputs



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Dynamic Operation

- The propagation delay (t_p) of a gate defines how quickly it responds to a change at its input(s)
- It is measured between the 50% transition points of the input and output waveforms
- Response times for rising or falling input waveforms are different:
  - t_{pLH}: response time for a low to high (positive) output transition
  - t_{pHL}: response time for a high to low (negative) output transition

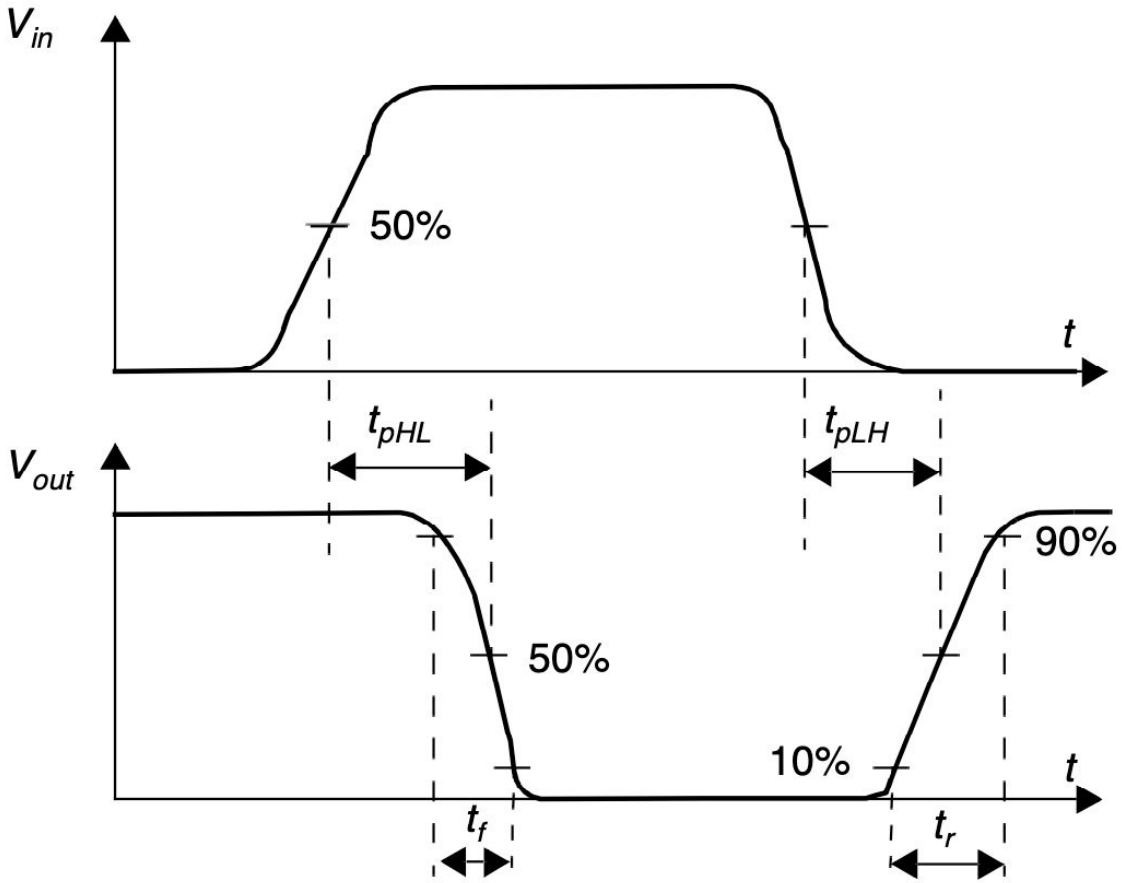
Rabaey, J. M., ***Digital Integrated Circuits: a Design Perspective***, 2nd ed., Prentice Hall, 2003.

Dynamic Operation

- t_p is defined as $(t_{pLH} + t_{pHL})/2$
- The rise time (t_r) and fall time (t_f) are metrics that apply to individual waveforms rather than gates and express how fast a signal transits between the different levels
- To avoid uncertainty, they are defined between the 10% and 90% points of the waveform
- They are affected by the strength of the driving gate and the load presented by the node

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Dynamic Operation



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